

Heteroencoding

Baseline Experiment

Testing heteroencoding with a synthetic dataset, consisting of random integers from 0 to 9 encoded as random SHRs with population 10. This circuit adds 30 random elements and randomly removes a few elements.

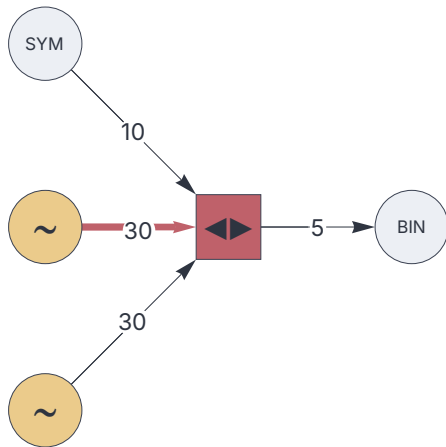
The heteroencoder component successfully clusters this data into 10 bins.

Configuration

```
<< "Circuits.m"
```

```
dataflow =  
  {  
    noise[][11][],  
    11 → {1000, 30},  
    noise[][12][],  
    12 → {1000, 30},  
    input[codec] [21] [],  
    21 → {1000, 10},  
    heteroencoder[] [101] [{11"%",12 "-%", 21"%"}],  
    output[binning] [] [101"%"],  
    101 → {500, 5}  
  };
```

```
c = Circuit[ dataflow]; c[Graph, ImageSize→ 240]
```



Synthetic data generation

```
dataset = RandomInteger[ {0, 9}, 10000];
```

Train

```
bins = c /@ dataset;
```

Analyze

```
SortBy[Tally[dataset], Last]
```

```
{{3, 938}, {1, 947}, {6, 983}, {2, 989}, {8, 995},  
 {5, 1011}, {9, 1022}, {4, 1027}, {0, 1033}, {7, 1055}}
```

```
SortBy[Tally[bins], Last]
```

```
{{1, 938}, {5, 947}, {6, 983}, {7, 989}, {9, 995},  
 {10, 1011}, {2, 1022}, {4, 1027}, {3, 1033}, {8, 1055}}
```

The algorithm has identified 10 clusters and correctly classified the data.